

An Overview of Optimization Methods for Home Energy Management Systems

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Abstract—Energy Management System (EMS) is a major component of a smart grid and significant for the operational qualification. Controlling the residential grid, resulting into improving cost, emission, and comfort. The study aims to investigate the optimization methods used in Home Energy Management System (HEMS) and evaluate their effectiveness. The paper discusses the architecture of HEMS, which is classified into three layers: physical, communication, and software. The physical layer comprises measurement systems and sensors for data acquisition, including smart meters, IoT sensors and smart appliances. The communication layer facilitates the interconnection between the control systems, central platforms, and smart devices. The software layer comprises the algorithmic and programming parts to optimize the system. Culminate into evaluating the different optimization methods used in HEMS, including mathematical, meta-heuristic, and artificial intelligence models.

Index Terms—Home Energy Management Systems, Artificial Intelligence, Optimization, Forecasting

I. INTRODUCTION

Currently, there is a growing demand for regulating and monitoring energy consumption and production within the residential sector. Home energy management systems (HEMS) are digital platforms that oversee and regulate energy production, storage, and generation within a household. These systems typically aim to achieve objectives, such as cost savings, maximizing self-sufficiency, and minimizing greenhouse gas emissions. As the use of electric vehicles, heating, residential generation plants, and variable pricing schemes has become more prevalent, HEMS are becoming increasingly popular owing to their potential to save energy and reduce costs.

In recent years, the use of energy management systems has increased, resulting in an upward trend in the research and development of their architecture. According to [1], the global market for Energy Management Systems (EMS) is currently estimated at 28.25 billion and is projected to experience substantial growth, reaching 82.34 billion by the year 2030.

This study was carried out in MESSI – Management Energy Systems for Smart Islands project, within the PRIN PROGETTI DI RICERCA DI RILEVANTE INTERESSE NAZIONALE – Bando 2022 and received funding from the European Union Next-GenerationEU (PIANO NAZIONALE DI RIPRESA E RESILIENZA - PNRR MISSIONE 4 COMPONENTE 2, INVESTIMENTO 1.1 –D.D. n. 104 del 02/02/2022, 2022HMYX2C). This manuscript reflects only the authors' views and opinions, neither the European Union nor the European Commission can be considered responsible for them.

The main architecture classification discussed in this paper is divided into three layers, physical, communication, and software. The components used at the physical level mainly comprise the measurement systems and sensors used to acquire data. They primarily consist of smart meters that monitor energy consumption and production, such as thermal and electrical energy. [2] The communication layer can be categorized into two subcategories, communication architecture and communication protocols. The first section analyzes the centralized and decentralized communication techniques, while the second describes the various protocols for data transfer. Last, the main components of a Home Energy Management System (HEMS) at the software layer consist of a User Interface, Energy Management Software, Data Processing and Analytics, Communication Middleware, Security Mechanisms, Automation Engine, API Integration, and Data Visualization. [3] Moreover, the concept of dynamic pricing has been proposed to smooth the unpredictability. It is considered an essential element in the architecture of a smart grid, and it is utilized to reduce the operational cost, but also improve the environmental impact. Some of the most known tariffs are Increasing-block pricing (IBP), Real-time Pricing (RTP), Time-of-Use (TOU), Critical Peak Pricing (CPP).

The key factors of a highly adjustable and replicable HEMS are the integration of the framework with common system architecture and semantic-level interoperability between the operational components. Some of the widely included in HEMS standards are EN 50491-12, OpenADR 2.0, SAREF, ISO 50001, NIST SG, IEC 62056, IEC 63110, and ISO 15118.

This study aims to investigate how optimization methods are implemented in Home Energy Management Systems (HEMS) and evaluate their effectiveness in reducing energy consumption and costs. Thus, gathering the aforementioned information leads to a comparison of the different optimization methods used in HEMS between mathematical, meta-heuristic, and artificial intelligence.

The remainder of this paper is organized as follows. Section II mentions the optimization techniques and algorithms of a HEMS, especially on the math-heuristic, meta-heuristic, and artificial intelligence models. Section III discusses the results of the optimization methods' analysis. Section IV concludes the paper and mentions future work.

II. OPTIMIZATION TECHNIQUES OF HEMS

Home energy management systems must possess six main features: energy consumption forecasting, energy production forecasting, load scheduling, control, monitoring, and user comfort. To achieve the aforementioned aspects, the utilization of optimization techniques and algorithms is necessary. There are several available models and approaches; however, in this study, as shown in Fig. 1, Mathematical, Meta-heuristic, and Artificial Intelligence are analyzed.

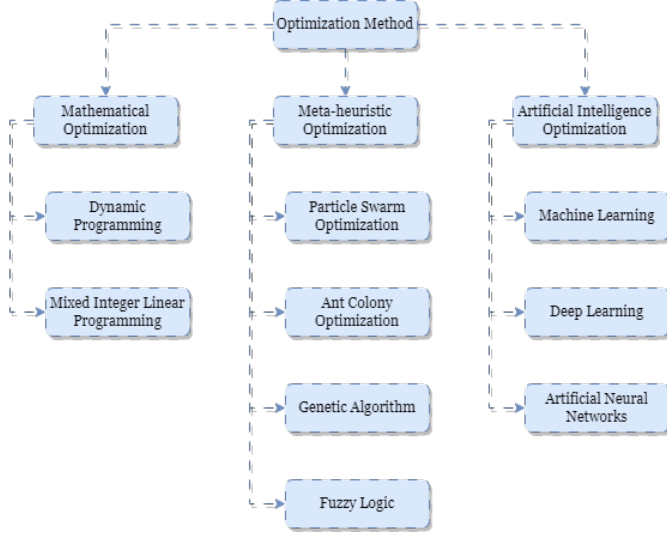


Fig. 1. Optimization techniques taxonomy

A. Mathematical Optimization Methods

Mathematical methods are based on a deterministic approach to a problem, which depends on the input values to achieve optimal scheduling. Nevertheless, the capabilities of the mathematical methods are limited since they are not suitable for multi-objective problems. The methods this chapter discusses are Mixed Integer Linear Programming and Dynamic Programming.

1) *Mixed Integer Linear Programming*: Mixed integer linear programming (MILP) is a powerful mathematical modeling technique that can effectively address complex optimization challenges and help identify trade-offs between conflicting objectives, as noted by [4]. This approach can enhance our understanding of energy management systems and empower the decision-making to optimize the grid.

Mathematical programming, particularly Mixed Integer Linear Programming (MILP), is widely explored in energy management systems due to its preciseness, flexibility, and extensive modeling abilities. [5]

In [6], the author utilized a MILP algorithm in a HEMS with the objective of minimizing the operating costs for the next 24 hours. The results of the developed HEMS model consistently outperform the experimental results in various case studies. Specifically, for the winter period and 2022 prices, the developed HEMS model achieves a cost reduction

of approximately 30% compared to the experimental system. In the summer months, the model performed more effectively with a 40% reduction. This significant real case study has proved several advantages, however, there are some limitations such as assumptions on the day-ahead electricity prices, lack of PV generation forecasting and given the knowledge of the appliances' power as known.

Similarly, in [7], the author conducted a performance analysis of a mixed-integer linear programming (MILP)-based energy management system. It formulated a linear model predictive controller (MPC) based on the modular modeling approach, cost values, and constraints of each prosumer and the connections between them. The results showed that the proposed EMS, when combined with supplementary system components, can achieve an annual cost-saving potential of 3-6%. Nonetheless, it is directly associated with extravagant tools such as MATLAB, rendering it hard to apply as poses challenges for its practical application due to factors such as cost, complexity, resource requirements, and dependency on specific software tools. .

2) *Dynamic Programming*: According to [8], this is a programming technique that solves an algorithmic problem by decomposing the primary problem into subproblems and then analyzing them to find the optimal general solution to the problem. This technique was first introduced by Bellman in 1962 [9], who developed a formula aimed at addressing the Traveling Salesman problem using dynamic programming.

In [10], the EMS optimized the response of energy demand and generation in the household based on electricity tariffs and supply conditions (for example, EV). By contrast, the study does not incorporate strategies for enabling the EMS to adjust to the specifications of the house and the management of an entire neighborhood in a multi-agent setting. Furthermore, additional devices and supplementary services such as regulation reserves can be integrated into the EMS.

In the study [11], the author introduced an EMS based on the DP algorithm for grid-connected residential PV systems equipped with battery energy storage systems. Its main purpose is to minimize the daily cost of electricity for residential customers by efficiently scheduling and managing the PV-BESS systems. The findings indicated that operating a BESS adaptably is advantageous and brings about positive outcomes, especially when dealing with fluctuating tariffs and homeowners who possess solar PV systems and plan to install BESSs can financially profit from transitioning to a TOU tariff. Nevertheless, this study is limited only to the Australian energy market, and it is only applicable to PV-BESS integrated houses without engaging in load optimization. These limitations could be resolved by integrating the existing system with residential load scheduling and controlling.

A different aspect of the DP algorithm utilization has been proposed in [12], in which thermal energy optimization is implemented. The leading objective of the study is to minimize consumption without affecting user comfort. The process is formulated by dynamic programming stochastic optimization. This optimization method has been compared with 3 different

scenarios, a no set-point and no storage unit scenario, a set-point but without storage unit scenario, and a no set-point but with storage unit scenario. The energy consumption reduction compared with each scenario was 21%, 19.2% and 3.04% respectively. Thus, the scenarios without thermal storage are less cost-efficient. Nevertheless, the current study is limited only to thermal energy optimization, without options for controlling, monitoring, and optimizing the entire residential grid.

B. Meta Heuristic Methods

Mathematical optimization methods are commonly utilized in modeling energy management systems for single objectives, such as optimizing for the optimal financial solution in the grid. Additionally, some problems involving the combination of different objectives are too complex to be solved, making it necessary for meta-heuristic algorithms to find acceptable solutions within a reasonable time frame. [13]

Some of the most used techniques in EMS are Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), Fuzzy Logic (FL), Genetic Algorithm (GA).

1) *Particle Swarm Optimization:* Particle Swarm Optimization (PSO) was introduced by Kennedy in 1995 and is grounded in social behaviors observed in animals such as birds or fish, exemplified in activities such as migration [14]. In PSO, particles navigate through the search space, seeking the optimal solution to a problem by sharing their positions, directions, and velocities among themselves [15]. To determine the optimum solution in PSO, particles move to their previous best positions in each iteration, resulting in the collective movement of the particle swarm toward the optimal position [16].

In [17], the author described the use of Particle Swarm Optimization (PSO) in the context of optimizing the operation of Battery Storage Systems (BSS) in Direct Current (DC) microgrids. The simulation revealed an average 2.95% cost reduction compared with the base scenario. However, the increased convergence time and the lack of consideration of the battery efficiency, self-discharge, and maximum charging cycles on the equation, distinguish it as limited.

Similarly, in [18], the author focused on presenting a modified PSO algorithm, proposing a dynamic penalty function, and analyzing the cost function with constraints. The algorithm aims to find the optimal battery controls of a community microgrid, considering time-varying electricity prices, stochastic energy sources, and power demand. A 3-case scenario simulation has been executed resulting 12% cost reduction comparing the proposed model with an original cost function. Nevertheless, this study examined this model in a simulated environment without an experimental case and the consideration of the battery's degradation.

2) *Ant Colony Optimization:* Ant Colony Optimization (ACO) is inspired by the observation of the social behavior of ants. As ants traverse their paths, they leave pheromones to share the optimal route for finding food with other members of their colonies [19]. The rationale behind ACO is that colony

members leave pheromones along their path as a form of indirect communication. The path is reinforced each time more members use it, with the aim of sharing the correct route with the rest of the colony. In this case, the ACO technique commences with the initialization of key parameters, such as the number of colony members, randomly arranging the members in the region. [15].

In [20], the author aims to optimize power control and comfort management in buildings through the use of optimization algorithms such as ant colony optimization (ACO). The objective is to minimize energy consumption while maintaining occupants' comfort levels through penalty incentives methods. The simulations indicate that the ACO-based system achieves a significant reduction in the total power consumption compared to the GA-based system. In addition, the proposed ACO-based optimization model improved the occupants' comfort index while consuming minimum energy. While the current paper examines multiple scenarios with many variables, the authors did not include crucial components such as EV chargers.

Another example of this optimization method is presented in [21], in which ACO is specifically applied to schedule the appliances in the microgrid, which includes fifteen homes with six appliances each. By using ACO, the paper aims to achieve better performance in terms of cost reduction and peak-to-average ratio (PAR) improvement compared to a system without any optimization technique. Nevertheless, the results focus only on cost and ignore the other functions.

3) *Genetic Algorithm Optimization:* The Genetic Algorithm (GA) is an optimization method based on the principles of natural evolution and selection, as derived from Darwinian theory. GA represents a set of potential solutions to an optimization problem. Its application in energy management involves the processes of selection, crossover, and mutation of algorithmic solutions in each generation, with the goal of selecting the optimal operation of the grid by reaching the final generation of solutions [22].

The GA process is comprised of three mechanisms: selection, crossover, and mutation. Selection occurs when the algorithm chooses suitable candidate solutions and transfers them to the set of solutions for the next generation. The crossover technique combines two candidate solutions and transfers them as one to the next generation. Finally, the mutation technique alters one candidate solution [23].

Additionally, due to mutations and the exchange of solutions from generation to generation, GA rarely falls into local optimization (local optima) but typically converges towards global optimization (global optima).

In [24], the author proposed a HEMS that was programmed based on genetic algorithm for load scheduling of the appliances in a household in Morocco. The system not only considers dynamic pricing of electricity but also the optimization for solar energy usage and user comfort. The results of this case study revealed that there are significant cost savings in each scenario without compromising the user's comfort level. Summarizing, in the first and third scenarios, the cost reduction

was apparent without compromising the user comfort. In the second scenario, there is a need for ancillary services, battery, grid, or expansion of the solar installation. Therefore, there are significant results in improving the yield and maximizing the comfort of the residents, however, the algorithm has not been tested in comparison to other algorithms.

In [25], the author proposed a household energy demand management strategy based on operating power using a genetic algorithm. The objective of this paper is cost minimization combining user response and power operation. The criteria of optimization are load response, dynamic pricing, and operating power. The results of the proposed technique were a 29% reduction in daily electricity cost, a peak-to-average ratio (PAR) reduction of 36.2%, an improvement in running speed, and authorization for users to choose equipment. Nevertheless, the authors have not considered criteria such as price incentives or the impact of decentralized energy resources.

4) *Fuzzy Logic Optimization*: Fuzzy Logic (FL) addresses uncertain problems where there is no knowledge of absolute correctness or incorrectness, mimicking the functioning of the human brain. To tackle such issues, a logic with multiple truth values is adopted, where truth lies between absolutely true and absolutely false.

In [26], the design and optimization of a fuzzy logic controller (FLC) for minimizing grid power profile fluctuations in a residential microgrid with wind and solar generation and a controllable battery load were described. The objective of this approach is to minimize fluctuations in the grid power profile while maintaining the battery state of charge within safe limits. After simulations of one year data, the system was tested experimentally in the microgrid for one month showing promising results. The criteria authors utilized to compare the proposed algorithm with the SMA strategy and NPT, were the maximum power value delivered by the grid (PGmax), the maximum power sold in the grid (PGmin), the maximum rate-of-change of the power profile (MPD), the average absolute mean rate-of-change of the power profile (APD), and the frequency deviation (PPV). According to the aforementioned criteria, the proposed FL model achieved better results on PGmax, PGmin and MPD than the other two alternatives. Nevertheless, slightly worse results were noted in PPV and APD.

Paper [27], discussed the development of an adaptable autonomous energy management solution for residential heating, ventilation, and air conditioning (HVAC) systems in smart grid environments. The authors utilize fuzzy logic controllers (FLCs) to design the smart thermostat, which allows for the control of temperatures without user interaction. The model learns based on user preferences and adjustments. Employing the aforementioned model and simulating it for one month, the results showed a noteworthy reduction in energy consumption by approximately 21.3%. This optimization case controls only the HVAC system and especially the main smart thermostat, without utilizing other aspects of a smart home, such as residential generation systems or appliances.

C. Artificial Intelligence

Artificial Intelligence is the engineering and science behind the process of making systems think and act rationally. It incorporates the power of the computer systems to train it and to approach issues in a human manner. Artificial intelligence methods are used in building energy systems for applications like energy use prediction, forecasting energy production heat transfer modeling, central plant control, and fault detection and diagnostics [28]. Artificial Intelligence is the main system, while Machine Learning is a subfield of it. Deep learning is a subset of ML and ANN is located on the core of this Venn Diagram.

The discussed techniques are Machine Learning, Deep Learning, Artificial Neural Networks.

1) *Machine Learning*: Machine learning (ML) is a field within artificial intelligence that focuses on developing algorithms to improve computer performance through experience, assisting in data analysis, and solving complex problems across various domains, such as energy management. [29]. Machine learning contains 3 subcategories, supervised, unsupervised, and semi-supervised. The main difference among the 3 categories is that supervised relies on labeled data, while unsupervised operates on unlabeled data. Semi-supervised learning lies between labeled and unlabeled data during data training.

In [30], the author aimed to curtail the household's total net daily cost by minimizing the electricity purchase expenses and maximizing the net revenues from electricity sales to the grid. Different types and complexity of forecasting algorithms, such as the General Regression Neural Network (GRNN) and Elman Neural Network (ENN), were examined. The comparison between GRNN and ENN has shown that both algorithm's impact acts positively on HEMS. Nevertheless, major differences have been in the results. GRNN optimization utilizes more RES power, 159.3 kWh resulting in an increased revenue, 14.6 €/day, due to higher power export to the grid, 66kWh/day. Additionally, higher State-of-Charge (SoC) levels have been observed in the EV during the day, providing readiness to the user. Based on the aforementioned data, the GRNN shows better results than ENN. However, the study is executed only at the simulation level, allowing further investigation on the experimental level.

2) *Artificial Neural Networks*: Artificial neural networks (ANNs) are an alternative technology that can predict energy consumption more reliably than traditional simulation models and regression techniques [31]. Artificial Neural Networks were inspired by biological neural networks and the functioning of the human brain, facilitating pattern recognition and problem-solving by computational systems. ANNs consist of a layer of nodes, which in turn includes two layers: the input layer and the output layer. Each node is characterized by two values: the weight and the threshold. The operation of ANNs depends on these values, as the output value of a node needs to exceed its threshold in order to be activated, allowing the data to be passed to the next layer. Otherwise, data are not transmitted to the next layer of the network.

In [32], the authors proposed a novel dynamic model for HEMS based on ANN and Reinforcement Learning. The main objectives are to reduce the electricity cost and grid overload without affecting user comfort. The results of the paper were that in a 12-hour time frame, it was observed a maximum cost reduction of 24%. On the other side, the load forecasting and the price forecasting were not so accurate with a 25% and 9.5% deviation, respectively. Thus, the main remarks are that it can be implemented in a wider share of residents and for a longer time frame. Moreover, the selection and execution of the algorithms could be modified for a more accurate result. Meanwhile, the implementation of the demonstrator smart plug is considered a reliable and low-cost solution, and it could be utilized for commercial uses.

Equivalently, the authors in [33] proposed a combination of HEMS and Bluetooth Low Energy technology for monitoring and controlling the residential grid. The current approach is based on an ANN that operates to forecast the energy requirements but also considers the residents' feedback. Moreover, a comparison between a standard IEEE 802.15.4 protocol and BLE protocol was executed, with the results indicating BLE is much more efficient. The results were promising but there is a margin for improvement. Components such as RES, BSS and more dynamic tariffs can be implemented to provide a more spherical and reliable solution.

3) *Deep Learning*: Deep learning (DL) is a machine learning technique using multiple layers of interconnected units to learn hierarchical representations of data, producing state-of-the-art results. It is an effective, supervised, time, and cost-efficient machine learning approach that can be applied to various complex problems, including energy optimization [34].

Deep reinforcement learning (DRL) works by using algorithms incorporating value function, policy, reward, model, exploration vs. exploitation, and representation to handle tasks [35].

In [36], the authors approached a residential energy management problem at a smart home with a power grid connection, energy storage system, and home loads, to minimize the electricity cost. The authors approached the topic with deep reinforcement learning, and it was formulated with a finite Markov decision process. The results show that the proposed model's accumulated electricity costs are reduced by 11.38% compared to the case without EMS, and it is close to the theoretical optimal cost. While the study reveals constructive results, it did not consider a more complex implementation with HVAC, EV, etc. Nonetheless, there is no comparison between any other algorithm to assess the efficiency.

In [37], the authors proposed an hour-ahead demand response algorithm to optimize the home energy management systems and to make optimal, real-time decisions for different home appliances in a decentralized manner using reinforcement learning (RL) and artificial neural networks. The results of the study show that the algorithm can effectively manage energy for multiple appliances, minimize energy bills, and dissatisfaction costs. To validate the superiority of the proposed model, the authors solve the same optimization problem

with the MILP approach and compare it with the ANN-RL model. The results were that the MILP approach offers an efficient and stable optimization. On the contrary, the ANN-RL approach in the first iterations does not operate well since it is in the training process, but in the long run, after 80 trials, outperforms the MILP model. Nevertheless, the authors did not consider energy usage and work time patterns neither the case of producing energy as a prosumer and self-using it or selling it back to the grid to maximize the economic benefits.

III. DISCUSSION

The main tasks of the aforementioned algorithms are to reduce the energy cost, reduce the emissions and increase the user comfort of the residents. These goals are achieved by upgrading the current residential grid to a smart grid, with options such as monitoring and controlling the appliances and the residential production plants. However, challenges and limitations have been observed, such as assumptions in case studies, inaccurate forecasting, wrong scheduling of devices, small-scale simulations, and insufficient objective functions for fully featured optimization. Evaluation of these algorithms is done via simulation and comparison between forecasted and real-time data to rate the efficiency of each algorithm.

In this overview, an analysis of 18 different approached case studies has been conducted and shown that the primary objectives of the current trend in HEMS are the minimization of cost, increase of profit, optimize the scheduling of the energy between storing and utilizing RES, minimizing power consumption. On the other hand, there is a lack of studies on power quality or power losses of the residential grid and only few research focuses on voltage fluctuations, fault diagnosis, harmonics control, and frequency stabilization, which are considered supplementary tasks of a HEMS. Therefore, the current trend of HEMS research aims at revenues and not on the quality of the grid, giving the potential to investigate it further as an extra function of an integrated HEMS.

Moreover, the studies that have the main objective the cost reduction have been gathered and compared as indicated in Fig. 2. Unfortunately, the assumptions, installations, and conditions of each case study are different. Therefore, the comparison does not provide some solid outcomes, but a general image of each algorithm's efficiency is demonstrated.

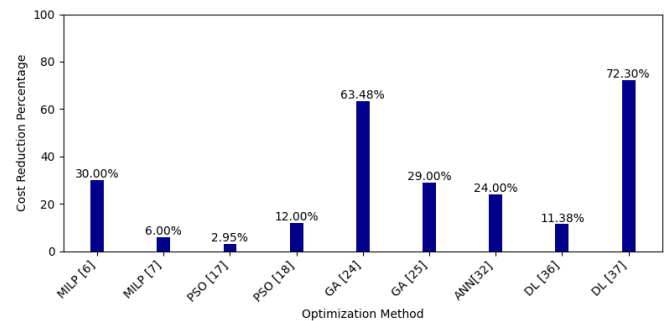


Fig. 2. Optimization methods comparison

IV. CONCLUSION

HEMS is an uprising technology, full of potential in the research and development side but also in the industrial and commercial sectors. The development of HEMS requires robust and efficient optimization techniques to encounter real-time goals and anomalies on the smart grid. While many research studies have been published, there are still gaps in the improvement of optimization of HEMS, such as algorithm efficiency, scalability, and multi-objective optimization. This paper conducts a comprehensive literature survey to examine the optimization techniques associated with energy management systems, address their limitations, and introduce our plans regarding the development of a novel algorithm to control smart homes. The main aspect of our research plan is the deployment and development of an advanced home energy management system based on an open-source software platform and time-series database, which will be integrated with artificial intelligence techniques for optimization of energy management and the improvement of comfort index and financial viability. The main features of our future proposal will be adaptive user behavior modeling, high scalability, and low cost and power consumption installation. Future works entail evaluating some algorithms through simulations and comparing them with our innovative approach.

REFERENCES

- [1] "https://www.fortunebusinessinsights.com/industry-reports/energy-management-system-market-101167."
- [2] J. Leitao, P. Gil, B. Ribeiro, and A. Cardoso, "A survey on home energy management," *IEEE Access*, vol. 8, pp. 5699–5722, 2020.
- [3] J. D. Wang, J. Q. Dai, Z. Q. Sun, and X. Y. Kong, "A software application for smart home energy management system," vol. 347–350, pp. 1079–1083, 2013.
- [4] M. Guo and N. Shah, *Bringing Non-energy Systems into a Bioenergy Value Chain Optimization Framework*, pp. 2351–2356, 2015.
- [5] C. A. Floudas and X. Lin, "Mixed integer linear programming in process scheduling: Modeling, algorithms, and applications," *Annals of Operations Research*, vol. 139, pp. 131–162, 10 2005.
- [6] I. L. Gomes, M. G. Ruano, and A. E. Ruano, "MILP-based model predictive control for home energy management systems: A real case study in Algarve, Portugal," *Energy and Buildings*, vol. 281, 2 2023.
- [7] A. Moser, D. Muschick, M. Gölls, P. Nageler, H. Schranzhofer, T. Mach, C. R. Tugores, I. Leusbrock, S. Stark, F. Lackner, and A. Hofer, "A MILP-based modular energy management system for urban multi-energy systems: Performance and sensitivity analysis," *Applied Energy*, vol. 261, 3 2020.
- [8] "https://www.spiceworks.com/tech/devops/articles/what-is-dynamic-programming/."
- [9] R. Bellman, "Dynamic programming treatment of the travelling salesman problem," *Journal of the ACM*, vol. 9, pp. 61–63, 1 1962.
- [10] I. P. . E. Society., I. of Electrical, and E. Engineers, *2011 IEEE PES Innovative Smart Grid Technologies : IEEE PES ISGT Asia 2011 Conference : Smarter Grid for Sustainable and Affordable Energy Future : November 13th-16th Perth, Australia*.
- [11] B. Jeddi, Y. Mishra, and G. Ledwich, "Fig. 1. schematic of residential rooftop pv-bess system dynamic programming based home energy management unit incorporating pvs and batteries," 2017.
- [12] R. M. Vignali, F. Borghesan, L. Piroddi, M. Strelec, and M. Prandini, "Energy management of a building cooling system with thermal storage: An approximate dynamic programming solution," *IEEE Transactions on Automation Science and Engineering*, vol. 14, pp. 619–633, 4 2017.
- [13] A. H. Gandomi, X.-S. Yang, S. Talatahari, and A. H. Alavi, "Metaheuristic algorithms in modeling and optimization," *Metaheuristic applications in structures and infrastructures*, vol. 1, pp. 1–24, 2013.
- [14] J. Kennedy and R. Eberhart, "Particle swarm optimization," pp. 1942–1948, IEEE.
- [15] Y. Zhao, Z. Zheng, and Y. Liu, "Survey on computational-intelligence-based UAV path planning," *Knowledge-Based Systems*, vol. 158, pp. 54–64, 10 2018.
- [16] Y. Zhang, S. Wang, and G. Ji, "A comprehensive survey on particle swarm optimization algorithm and its applications," *Mathematical Problems in Engineering*, vol. 2015, pp. 1–38, 2015.
- [17] L. F. Grisales-Noreña, O. D. Montoya, and C. A. Ramos-Paja, "An energy management system for optimal operation of BSS in DC distributed generation environments based on a parallel PSO algorithm," *Journal of Energy Storage*, vol. 29, 6 2020.
- [18] M. A. Hossain, H. R. Pota, S. Squartini, and A. F. Abdou, "Modified PSO algorithm for real-time energy management in grid-connected microgrids," *Renewable Energy*, vol. 136, pp. 746–757, 6 2019.
- [19] M. Dorigo, M. Birattari, and T. Stutzle, "Ant colony optimization," *IEEE Computational Intelligence Magazine*, vol. 1, pp. 28–39, 11 2006.
- [20] I. Fatima, A. Khalid, S. Zahoor, A. Yasmeen, S. Arif, U. Zafar, and N. Javaid, *Home energy management system using ant colony optimization technique in microgrid*, vol. 12, pp. 267–279. Springer Science and Business Media Deutschland GmbH, 2018.
- [21] A. Galbazar, S. Ali, and D. H. Kim, "Optimization approach for energy saving and comfortable space using ACO in building," *International Journal of Smart Home*, vol. 10, pp. 47–56, 2016.
- [22] M. A. Albadr, S. Tiun, M. Ayob, and F. AL-Dhief, "Genetic algorithm based on natural selection theory for optimization problems," *Symmetry*, vol. 12, p. 1758, 10 2020.
- [23] J. Tu and S. Yang, "Genetic algorithm based path planning for a mobile robot," pp. 1221–1226, IEEE.
- [24] R. E. Makroum, A. Khallaayoun, R. Lghoul, K. Mehta, and W. Zörner, "Home energy management system based on genetic algorithm for load scheduling: A case study based on real life consumption data," *Energies*, vol. 16, 3 2023.
- [25] X. Jiang and C. Xiao, "Household energy demand management strategy based on operating power by genetic algorithm," *IEEE Access*, vol. 7, pp. 96414–96423, 2019.
- [26] D. Arcos-Aviles, J. Pascual, L. Marroyo, P. Sanchis, and F. Guinjoan, "Fuzzy logic-based energy management system design for residential grid-connected microgrids," *IEEE Transactions on Smart Grid*, vol. 9, pp. 530–543, 2018.
- [27] A. Keshkar and S. Arzanpour, "An adaptive fuzzy logic system for residential energy management in smart grid environments," *Applied Energy*, vol. 186, pp. 68–81, 2017.
- [28] M. Krarti, "An overview of artificial intelligence-based methods for building energy systems," *Journal of Solar Energy Engineering*, vol. 125, pp. 331–342, 8 2003.
- [29] M. I. Jordan and T. M. Mitchell, "Machine learning: Trends, perspectives, and prospects," *Science*, vol. 349, pp. 255–260, 7 2015.
- [30] N. Koltsaklis, I. Panapakidis, G. Christoforidis, and J. Knápek, "Smart home energy management processes support through machine learning algorithms," *Energy Reports*, vol. 8, pp. 1–6, 6 2022.
- [31] H. S. Stern, "Neural networks in applied statistics," *Technometrics*, vol. 38, p. 205, 8 1996.
- [32] N. Ashenov, M. Myrzaliyeva, M. Mussakhanova, and H. S. Nunna, "Dynamic cloud and ANN based home energy management system for end-users with smart-plugs and PV generation," Institute of Electrical and Electronics Engineers Inc., 2 2021.
- [33] M. Collotta and G. Pau, "An innovative approach for forecasting of energy requirements to improve a smart home management system based on BLE," *IEEE Transactions on Green Communications and Networking*, vol. 1, pp. 112–120, 3 2017.
- [34] S. Dargan, M. Kumar, M. R. Ayyagari, and G. Kumar, "A survey of deep learning and its applications: A new paradigm to machine learning," *Archives of Computational Methods in Engineering*, vol. 27, pp. 1071–1092, 9 2020.
- [35] P. Hernandez-Leal, B. Kartal, and M. E. Taylor, "A survey and critique of multiagent deep reinforcement learning," *Autonomous Agents and Multi-Agent Systems*, vol. 33, pp. 750–797, 11 2019.
- [36] I. C. I. Society, I. N. N. Society, I. of Electrical, and E. Engineers, *2018 International Joint Conference on Neural Networks (IJCNN) : 2018 proceedings*.
- [37] R. Lu, S. H. Hong, and M. Yu, "Demand response for home energy management using reinforcement learning and artificial neural network," *IEEE Transactions on Smart Grid*, vol. 10, pp. 6629–6639, 11 2019.